LAB PYTHON PROGRAMS
CSA1718-ARTIFICIAL INTELLIGENCE

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5. Write a python program for Missionaries and Cannibal Problem

PROGRAM:

```
from collections import deque

def solve():
 start = (3, 3, 0)
 goal = (0, 0, 1)
 moves = [(1, 0), (2, 0), (0, 1),
 (0, 2), (1, 1)]

 queue = deque([(start, [])])
 visited = {start}

 while queue:
 state, path = queue.popleft()
 m, c, boat = state

 if state == goal:
 for step in path + [state]:
 print(step)
 return

 for dm, dc in moves:
 if boat == 0:
 new = (m-dm, c-dc, 1)
 else:
 new = (m+dm, c+dc, 0)

 nm, nc, _ = new

 if not (0 <= nm <= 3 and 0 <= nc <= 3):
```

```
continue

if ((nm > 0 and nm < nc) or
(3-nm > 0 and 3-nm < 3-nc)):
continue

if new not in visited:
visited.add(new)
queue.append((new, path + [state]))

solve()
```

OUTPUT

```
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
Goal Reached!
```